

Institute/Department	UNIVERSITY INSTITUTE OF ENGINEERING (UIE)	Program	Bachelor of Engineering (Hons.) (Computer Science and Engineering) (with Specialization in Artificial Intelligence and Machine Learning)(CS221)
Master Subject Coordinator Name:	Amandeep Kaur Sekhon	Master Subject Coordinator E-Code:	E19849
Course Name	Full Stack -II	Course Code	24CSP-337

Lecture	Tutorial	Practical	Self Study	Skilling	TC	TGT	TGP	Studio	Credit	Subject Type
0	0	4	0	0	0	0	0	0	2.00	P

Course Type	Course Category	Mode of Assessment	Mode of Delivery
Minor Core	Graded (GR)	Practical Examination (PRAC)	Practical (PRAC)

Mission of the Department	M1: To provide relevant, rigorous and contemporary curriculum and aligned assessment system to ensure effective learning outcomes for engineering technologies. M2: To provide platform for industry engagement aimed at providing hands-on training on advanced technological and business skills to our students. M3: To provide opportunities for collaborative, interdisciplinary and cutting-edge research aimed at developing solutions to real life problems M4: To imbibe quest for innovation, continuous learning and zeal to pursue excellence through hard work and problem-solving approach M5: To foster skills of leadership, management, communication, team spirit and strong professional ethics in all academic and societal endeavours of our students
Vision of the Department	To be recognized as a centre of excellence for Computer Science & Engineering education and research, through effective teaching practices, hands-on training on cutting edge computing technologies and excellence in innovation, for creating globally aware competent professionals with strong work ethics whom would be proficient in implementing modern technology solutions and shall have entrepreneurial zeal to solve problems of organizations and society at large.

Program Educational Objectives(PEOs)

PEO1	To be able to explore areas of research, technology application & innovation and make a positive impact in different types of institutional settings such as corporate entities, government bodies, NGOs, inter-government organizations, & start-ups.
PEO2	To be able to design, and implement technology and computing solutions to organizational problems, effectively deploy knowledge of engineering principles, demonstrate critical thinking skills & make the intellectual connections between quantitative and qualitative tools, theories, and context to solve the organizational problems
PEO3	To be able to work with, lead & engage big and small teams comprising diverse people in terms of gender, nationality, region, language, culture & beliefs. To understand stated and unstated differences of views, beliefs & customs in diverse & interdisciplinary team settings
PEO4	To be able to continuously learn and update one's knowledge, engage in lifelong learning habits and acquire latest knowledge to perform in current work settings
PEO5	To continuously strive for justice, ethics, equality, honesty, and integrity both in personal and professional pursuits. Able to understand and conduct in a way that is responsible and respectful.

Program Specific OutComes(PSOs)

PSO1	PSO1: Graduates will be able to analyze, design, and develop intelligent systems and applications by applying core concepts of Artificial Intelligence and Machine Learning across diverse domains.
PSO2	PSO2: Graduates will demonstrate proficiency in utilizing advanced AI/ML tools, frameworks, and technologies to innovate, implement, and manage projects in the rapidly evolving field of Artificial Intelligence and its allied application areas.
PSO3	PSO3: Graduates will apply AI, Machine Learning, and Data Analytics techniques to address real-world challenges, delivering effective and ethical solutions for industry, research, and societal needs.

Program OutComes(POs)	
PO1	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO2	Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO3	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety and the cultural, societal, and environmental considerations.
PO4	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data and synthesis of the information to provide valid conclusions.
PO5	Create, select, and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO6	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context to technological change.
PO13	Demonstrate the capability to apply analytic thought to a body of knowledge, including the analysis and evaluation of policies, and practices. Identify relevant assumptions or implications, logical flaws and loopholes in the presented arguments.
PO14	Demonstrate to create, perform, or think in different and diverse ways about the given scenario. Innovate and perform tasks in a better manner, view a problem or a situation from multiple perspectives, think 'out of the box' and generate solutions to complex problems in unfamiliar contexts
PO15	Demonstrate the ability to identify with or understand the perspective, experiences, or points of view of another individual or group, and to identify and understand other people's emotions
PO16	Demonstrate the ability to participate in community-engaged services/ activities for promoting the well-being of society
PO17	Demonstrate the acquisition of knowledge of the values and beliefs of multiple cultures, capability to effectively engage in a multicultural group/society and interact respectfully with diverse groups and gender sensitivity and adopting a gender-neutral approach, as also empathy for the less advantaged and the differently-abled including those with learning disabilities.

Text Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Full Stack Development with Spring Boot and React	Juha Hinkula	4th Edition	Packt Publishing	2023
2	Learning React	Alex Banks, Eve Porcello	2nd Edition	O'Reilly Media	2020
3	Spring Boot Up and Running	Mark Heckler	1st Edition	O'Reilly Media	2021

Reference Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Designing Data-Intensive Applications	Martin Kleppmann	1st Edition	O'Reilly Media	2017
2	Docker Deep Dive	Nigel Poulton	1st Edition	Leanpub	2017
3	High Performance Java Persistence	Vlad Mihalcea	1st Edition	Vlad Mihalcea	2016
4	Learning Java	Marc Loy, Patrick Niemeyer	5th Edition	O'Reilly Media	2020

5	Spring Boot Security in Action	Laurentiu Spilca	1st Edition	Manning Publications	2020
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Course OutCome	
SrNo	OutCome
CO1	Recall fundamental concepts, terminologies, and components of full-stack development, including client-server architecture, REST APIs, state management, authentication, event-driven systems, and deplo
CO2	Explain the architecture and workflow of a full-stack social media management platform, including data flow, component interaction, authentication mechanisms, and system integration.
CO3	Apply full-stack development principles to implement functional modules such as post creation, state management, authentication, and API integration with proper validation and error handling.
CO4	Analyze system behavior and performance across integrated modules to identify bottlenecks, security issues, and optimization opportunities in data handling, rendering, and communication.
CO5	Evaluate design choices, architectural patterns, and implementation strategies for scalability, reliability, and maintainability, including caching, event-driven processing, and access control mechani
CO6	Design and develop a complete, production-ready full-stack social media management system integrating frontend, backend, authentication, analytics, scheduling, and CI/CD pipelines.

Lecture Plan Preview-Practical					
Unit No	SessionNo	Experiment Name	Text/ Reference Books	Pedagogical Tool**	CO - BT Level Mapping
1	1	Post Composer UI with Platform Constraints	T-Learning React	Hand On Activity based	CO1 - BT1; CO3 - BT3
1	2	Draft Management & API Integration	,T-Full Stack Development with Sp,T-Learning React,R-Designing Data-Intensive Appli	Hand On Activity based	CO2 - BT2; CO3 - BT3
1	3	Redux Toolkit Setup & State Design	,T-Learning React,R-Designing Data-Intensive Appli	Hand On Activity based	CO1 - BT1; CO2 - BT2; CO3 - BT3
1	4	Optimized Selectors & Performance Handling	,T-Learning React,R-Designing Data-Intensive Appli	Hand On Activity based	CO1 - BT1; CO2 - BT2; CO3 - BT3; CO4 - BT4; CO5 - BT5; CO6 - BT6
1	5	JWT-Based Authentication System	,T-Full Stack Development with Sp,T-Learning React,R-Designing Data-Intensive Appli	Hand On Activity based	CO1 - BT1; CO2 - BT2; CO3 - BT3
1	6	Role-Based Access & Route Protection	,T-Full Stack Development with Sp,T-Learning React,R-Designing Data-Intensive Appli	Hand On Activity based	CO2 - BT2; CO3 - BT3
1	7	Interactive Calendar for Post Scheduling	,T-Full Stack Development with Sp,T-Learning React,R-Designing Data-Intensive Appli	Hand On Activity based	CO3 - BT3
1	8	Performance Optimization & Testing	,T-Full Stack Development with Sp,R-Designing Data-Intensive Appli	Hand On Activity based	CO4 - BT4; CO5 - BT5
2	9	REST API Design & Implementation	,T-Spring Boot Up and Running,R-High Performance Java Persiste,R-Learning Java,R-Spring Boot Security in Action	Hand On Activity based	CO1 - BT1; CO3 - BT3
2	10	Global Exception Handling & Logging	,T-Spring Boot Up and Running,R-Learning Java,R-Spring Boot Security in Action	Hand On Activity based	CO3 - BT3
2	11	Pagination & Sorting Implementation	,T-Spring Boot Up and Running,R-Designing Data-Intensive Appli,R-High Performance Java Persiste	Hand On Activity based	CO3 - BT3; CO4 - BT4
2	12	Caching & Query Optimization	,T-Spring Boot Up and Running,R-Designing Data-Intensive Appli,R-High Performance Java Persiste,R-Spring Boot Security in Action	Hand On Activity based	CO4 - BT4; CO5 - BT5

2	13	Secure APIs with JWT & RBAC	,T-Spring Boot Up and Running,R-Designing Data-Intensive Appli,R-Spring Boot Security in Action	Hand On Activity based	CO2 - BT2; CO3 - BT3; CO4 - BT4
2	14	Encryption & Secure Credential Storage	,T-Full Stack Development with Sp,T-Spring Boot Up and Running,R-Designing Data-Intensive Appli,R-Spring Boot Security in Action	Hand On Activity based	CO3 - BT3; CO4 - BT4; CO5 - BT5; CO6 - BT6
3	15	Kafka Producer-Consumer Implementation	,T-Full Stack Development with Sp,R-Docker Deep Dive	Hand On Activity based	CO2 - BT2; CO3 - BT3; CO5 - BT5
3	16	Reliability: Retry, DLQ & Idempotency	,T-Full Stack Development with Sp,R-Docker Deep Dive	Hand On Activity based	CO5 - BT5; CO6 - BT6
3	17	Continuous Integration Pipeline Setup	,T-Full Stack Development with Sp,R-Docker Deep Dive	Hand On Activity based	CO6 - BT6
3	18	Continuous Deployment & Release Strategies	,T-Full Stack Development with Sp,R-Docker Deep Dive	Hand On Activity based	CO6 - BT6
3	19	Full-Stack System Integration	,T-Full Stack Development with Sp,R-Designing Data-Intensive Appli,R-Spring Boot Security in Action	Hand On Activity based	CO5 - BT5; CO6 - BT6
3	20	Capstone Project	,T-Full Stack Development with Sp,R-Docker Deep Dive,R-Spring Boot Security in Action	Hand On Activity based	CO6 - BT6

Assessment Model			
Sr No	Exam Name	Max Marks	Weighted Marks
1	External Viva / Voce	40	40
2	Experiment-1	30	4.5
3	Experiment-2	30	4.5
4	Experiment-3	30	4.5
5	Experiment-4	30	4.5
6	Experiment-5	30	4.5
7	Experiment-6	30	4.5
8	Experiment-7	30	4.5
9	Experiment-8	30	4.5
10	Experiment-9	30	4.5
11	Experiment-10	30	4.5
12	Mid-Term Test	15	15

CO vs PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15
CO1	2	1	NA	NA	1	NA	NA	NA	NA	NA	NA	1	1	NA	NA
CO2	2	2	1	1	2	NA	NA	NA	1	1	1	1	2	1	NA
CO3	3	2	2	1	3	1	NA	1	1	2	2	1	2	2	NA
CO4	3	3	2	2	3	1	NA	1	1	2	2	1	3	2	NA
CO5	3	3	3	2	3	2	1	2	1	2	2	2	3	3	NA
CO6	3	3	3	3	3	2	1	2	2	3	3	2	3	3	1
Target	2.67	2.33	2.2	1.8	2.5	1.5	1	1.5	1.2	2	2	1.33	2.33	2.2	1

PO16	PO17	PSO1	PSO2	PSO3
NA	NA	1	1	1
NA	NA	1	2	1
NA	NA	2	3	2
NA	NA	2	3	2
NA	NA	3	3	3
1	1	3	3	3
1	1	2	2.5	2

